(i) $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

$$xy\frac{dy}{dx} + y^2 - 2x^2 = 0$$

$$x(xv)\left(v + x\frac{dv}{dx}\right) + x^2v^2 - 2x^2 = 0$$

$$x^2v^2 + x^3v\frac{dv}{dx} + x^2v^2 - 2x^2 = 0$$

$$\implies xv\frac{dv}{dx} + 2v^2 - 2 = 0$$

$$xv\frac{dv}{dx} + 2v^2 - 2 = 0$$

$$xv\frac{dv}{dx} = 2(1 - v^2)$$

$$\frac{v}{1 - v^2}\frac{dv}{dx} = \frac{2}{x}$$

$$\int \frac{v}{1 - v^2}\,dv = \int \frac{2}{x}\,dx$$

$$-\frac{1}{2}\ln|1 - v^2| = 2\ln|x| + C$$

$$\ln|x^4(1 - v^2)| = C$$

$$x^4(v^2 - 1) = C$$

$$x^4\left(\frac{y^2}{x^2} - 1\right) = C$$

$$\implies x^2(y^2 - x^2) = C$$

(ii) Let $y = xv$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

$$y\frac{dy}{dx} + 6x + 5y = 0$$

$$xv\left(v + x\frac{dv}{dx}\right) + 6x + 5xv = 0$$

$$xv\frac{dv}{dx} + (v + 2)(v + 3) = 0$$

$$\int \frac{v}{(v+2)(v+3)}\,dv = -\int \frac{1}{x}\,dx$$

$$\int \left(-\frac{2}{v+2} + \frac{3}{v+3}\right)\,dv = -\ln|x| + C$$

$$-2\ln|v+2| + 3\ln|v+3| = -\ln|x| + C$$

$$\ln\left|x\frac{(v+3)^3}{(v+2)^2}\right| = C$$

$$\frac{(y+3x)^3}{(y+2x)^2} = C$$

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Last updated: July 1, 2026